

QINGHAN JIA

Robot Learning · Contact-Rich Manipulation
Multimodal Data Collection & Processing (UMI)
AI for CAD · Robot Co-Design

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PROFILE

Working on robot learning for contact-rich manipulation, hands-on from hardware and data collection to model training and real-robot deployment. Interested in AI for CAD and hardware–policy co-design, where task data shapes both a robot's body and its behavior.

EDUCATION

Southeast University

M.S. in Systems Science — 2024 – Present

B.S. in Mathematics and Applied Mathematics GPA 3.58 / 4.00 — 2020 – 2024

EXPERIENCE

Midea AI Research Institute
2026
Research Intern

ContactWAM: Force–Tactile Data to Contact-Predictive World Action Model

- Debugged the end-to-end collection pipeline on **ContactUMI**, a handheld gripper streaming synced vision, tactile, wrench and pose.
- Built the data pipeline (time alignment, relative-pose actions, tactile depth/shear maps, LeRobot export) plus real-time IK and multi-stream visualizers.
- Proposed **ContactWAM**, a world action model whose contact expert predicts future touch and wrench with video and actions.
- Reached **77.5%** mean success on four real-robot contact-rich tasks, vs. 43.8% for $\pi_{0.5}$, 51.3% for tactile-pretrained FTP-1 and 45.0% for vision-only FastWAM.

Household Bimanual VLA: UMI Data-to-Deployment Pipeline

- Built a robot-free UMI collection pipeline with HTC Vive pose tracking, real-time quality checks, IK alerts and Rerun replay.
- Studied frames, action representations, cameras and data scale; ~7 h of UMI data reached **10/13** successes in a 6-minute real-robot run.

Shanghai Jiao Tong University
Summer 2025
Research Intern
Prof. Cewu Lu's Lab

Demo2Arm: Human Demonstrations to Task-Specific Robot Arms

- Built a differentiable co-design framework that turns demonstrated trajectories and six-axis wrenches into task-specific arm designs.
- Linked cage-deformed link geometry to differentiable FEA and rigid-body dynamics, so compliance and torque gradients shape the same links.
- Searched topology and geometry hierarchically; cut link deflection by **17.2%** and peak torque utilization by **16.2%**, with transfer to an unseen task.

PUBLICATIONS

Under Review
ICRA 2027

ContactWAM: From Handheld Force–Tactile Demonstrations to Contact-Predictive World Action Model

Qinghan Jia (First Author), et al. · contactwam.github.io

Under Review
ICRA 2027

Demo2Arm: From Human Demonstrations to Task-Specific Robotic Arms via Differentiable Co-Design

Qinghan Jia (First Author), et al. · demo2arm.github.io

SKILLS

Robot Learning	VLA & world action models · π_0 / $\pi_{0.5}$ fine-tuning · PyTorch
Data & Training	UMI & force–tactile data · multi-node pretraining (16 × 8 H20)
Robotics & Design	ROS 2 · Flexiv · ARX · MuJoCo · SolidWorks · differentiable FEA
Engineering	Python · C++ · Linux · Git · Rerun · LaTeX · Claude Code / Codex